November 2004

Issue # 11

LE RAYON DE SOLEIL Online

The Synchrotron SOLEIL newspaper

LUCIA is operational at SLS; the LINAC is being installed; the first building has been delivered at SOLEIL!

Now we are entering the home straight! At the same time as the "Sources" group sees the first LINAC components (provided by Thalès) arrive, they are installing the girders and magnets of the LT1 (the transfer line guiding the electrons from the LINAC to the Booster synchrotron). Just a few metres away, the construction companies are busy finishing the civil engineering work, the second phase of the synchrotron building, and installing the necessary infrastructure. It is easy to imagine the excitement which reigns at the construction site (as well as in our offices)!
The detailed source and accelerator programs are presented in this issue of "Rayon de SOLEIL", together with our approach to beamline control and data acquisition systems. The description of 3 hard X-ray beamlines dedicated to structural studies can also be found here, along with summaries of some of the scientific meetings which were initiated by SOLEIL during recent months. October saw us contribute to the national day of celebration of science and scientific culture (the 'fête de la science'), an occasion which has a high priority for us, evidenced by our participation in several scientific festivals and programs throughout the year.

On the SOLEIL building site the last Douglas pine claddings decorating the technical buildings, the central office building and the restaurant are being installed. The local population can already get a good idea of what the final site will look like since the first building – the 'Reception building and Visitor centre'- has now been "delivered" by the constructors. It will be opened to the public in December, after the security authorisation necessary for an ERP (establishment open to the public) has been received. This centre will welcome organised visits from the general public, and provide information both about SOLEIL and synchrotron radiation science in general.
Concerning the building of the synchrotron, the building is now "water tight" (roofing and windows installed) with the "ears" (the peripheral circle of laboratories and offices) now being installed. At this time the major activity has moved indoors, within the tunnels of the LINAC, the Booster synchrotron and the storage ring, and also at the technical sites which comprise, at the heart of the building, the necessary services for the operation of the source and beamlines. At the time these lines are being written it seems very likely that the central building, comprising the offices and the restaurant, will be delivered according to schedule (February 2005). We are, however, more concerned about the progress of the synchrotron and technical buildings, since the present state of their construction reveals a delay of 3 months (compared to the original timeline approved by the constructors). This delay is a subject of concern as it causes a knock on delay in the commissioning of the ring and installation of the

Aerial views of SOLEIL, November 2004

beamlines and we are trying to reduce it with all means at our disposal.
On the other hand, the different elements of the "source and accelerator" programme remain within the original timeline and meet the design specifications. It is therefore appropriate in this issue to focus on the current installation of the LINAC and the imminent installation of the Booster synchrotron, the characterisation of the magnets and the design of the vacuum chambers for the storage ring. Even though the delivery of certain components are on the "critical path", we

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expect that all items will be on site and tested in time to complete the storage ring installation by the end of the second quarter of 2005. It is therefore the right time to present, in this issue, our approach to the accelerator and beamline control and data acquisition systems - an essential element of the experimental programme.

The experimental division is currently working on several projects in parallel: the construction of the first 11 "Phase 1" beamlines, the design studies for the first lines of "Phase 2", as well as establishing the scientific environment at SOLEIL. Day to day tasks include finalising the design of critical components, procurement of equipment and the recruitment of staff for the first beamlines. The scientific prospects of the source are highlighted by the organisation of several workshops in various subject areas (some of which are described elsewhere in this issue), and by the setting up of numerous collaborations with teams from local laboratories and universities (and even some further afield). It is in this spirit that the framework agreements between SOLEIL and several universities and laboratories have been drafted and signed. We regard the development of such collaborations as essential for the future scientific success of SOLEIL.

In this edition of "Rayon de SOLEIL", we have chosen to focus on DIFFABS, ODE and PROXIMA1, three hard X-ray beamlines dedicated to X-ray diffraction and absorption spectroscopy in the domains of material sciences and biology. These three will be amongst the first beamlines to open for users in 2006. The highlighted beamlines may have very different domains of scientific interest, sample environment and beam characteristics, but all bear witness to the ambitious and cross disciplinary approach to the scientific programme at SOLEIL.

A last word on LUCIA: the X-ray micro-spectroscopy beamline installed, and now in operation, at the SLS (Swiss Light Source) in collaboration with the Swiss synchrotron. The beamline, which has already welcomed its first French and Swiss users, will be transferred to SOLEIL at the end of 2007. Up to then, the SLS programme committee will schedule beamtime access every 6 months, with a quota of beam reserved for the French user community. The first elemental distribution maps of soil samples have already been measured by scanning X-ray fluorescence spectroscopy with a spatial resolution of the order of 12x12 µm² (~1 µm spatial resolution is expected to be achieved soon). In collaboration with beamline staff, a team from the LPMC has already recorded a "first" in working out how to make useful X-ray absorption measurements under high pressure (12 GPa) at energies as low as the Phosphorous K edge, hence opening up new perspectives in the domain of Earth sciences. We offer our congratulations to all those, Swiss and French, who have contributed to the success of this project.

Denis Raoux
General Director

COMING SOON…

- 13-15 Decembre 2004: Workshop "Future directions in synchrotron radiation bio-crystallography"
  http://www.synchrotron-soleil.fr/workshops/bioxtal2004/
  (Synchrotron SOLEIL)
- 14-18 December 2004: European training school "New lights on ancient materials", with the support of the COST-G8 network.
  (Synchrotron SOLEIL)
- 21-22 March 2005: Workshop **"**New trends in gas phase VUV/soft X-ray high resolution spectroscopies at SOLEIL**"**.
  http://www.synchrotron-soleil.fr/workshops/high-resolution_march2005/
  Université Paris-Sud (Orsay)
  Amphithéâtre “Pierre Lehmann” – LAL, Bât. 200
- 19-22 April 2005: GMPCA Archaemetry Symposium with a session devoted to the applications of synchrotron radiation to archaemetry, organized in partnership with the synchrotron SOLEIL (INSTN, Saclay)

The signature of the convention between SOLEIL and the University of Paris VI

On the 8th September 2004, a framework agreement was signed between the University Pierre and Marie Curie (Paris VI) and SOLEIL, with the objective of fixing the general conditions of collaboration between SOLEIL and the UPMC. The signatories were the president of the UPMC, M. Gilbert Béréziat, and the director general of SOLEIL, M. Denis Raoux.

LE RAYON DE SOLEIL is a publication of the synchrotron SOLEIL civil company
Publication director : Denis Raoux
Rédactrice en chef : Marie-Pauline Gacoin
Coordination : Dominique Chandesris, Marie-Paule Level, Isabelle Quinkal
ISSN : 1767-4824 – Legal copyright November 2004

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Pictures of SOLEIL

To follow the building, connect to our webcam : www.synchrotron-soleil.fr

June 2004 - laying of the synchrotron roof structure.

22 June 2004 - opening of LUCIA, first SOLEIL beam-line, on the Swiss synchrotron SLS.

July 2004 - overall aerial views of the synchrotron SOLEIL and its buildings.

30 July 2004 - reception building opened to the public in December.

18-19 September 2004 - a large audience gathers around the SOLEIL model and display panels while visiting the exhibit "Lights on Heritage" (on the premises of the C2RMF - Louvre).

23 September 2004 - piping leading to T3 at the exit of the 6 air-cooling towers.

SOURCES AND ACCELERATORS PROGRAM

Activity is going on with ever greater intensity: all important orders have been placed, the follow up of in plant manufacturing by the SOLEIL teams is in full swing, planning of the installation in close connection with the progress of the buildings requires significant efforts and the preparation for the commissioning of the beams on the various accelerators is already underway.

Installation of the LINAC, of the LT1 transfer line, and the first Booster girder

What about HELIOS LINAC ?

The linear accelerator, or LINAC, is the first synchrotron "link" (figure 1). It has just been delivered at SOLEIL.

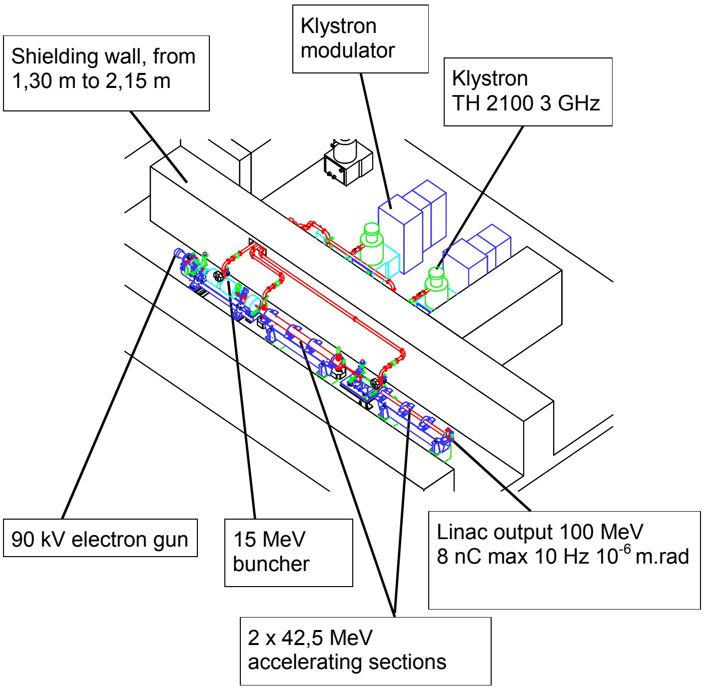


Figure 1: the main LINAC elements are the electron gun, the buncher and the accelerating sections

THALES delivered the two accelerating sections (figure 3), the buncher (figure 4), along with the girder used to join LT1, located in the hall.

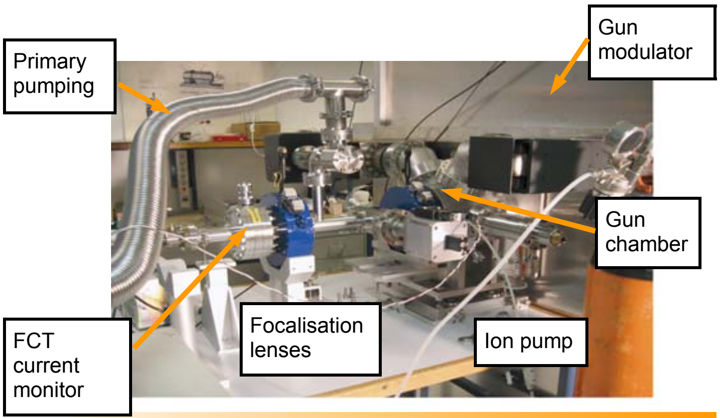


Figure 2: the front end of the LINAC: the gun, the ionic pumps, the beamline complete with the target to be measured

LINAC and Booster: who is building what?

- LINAC

Front end of the machine: *THALES communications*
Klystron modulators, general control-command of the machine: *EuroMeV*
Hydraulic equipment control-command: *THALES juret*
Buncher: *PMB*

- Booster

37 dipoles: *SIGMAPHI*
45 quadrupoles: *BINP*
28 sextupoles + 44 correctors: *SEF*
Dipoles and quadrupoles power supplies: *BRUKER*

Figure 3: Placing of one of the two accelerating sections (in the tunnel)

The general control-command of the machine and the hydraulic equipment remain to be completed. From the beneficial occupancy of the LINAC buildings in October, the THALES sub-contractors installed the hydraulic network and electrical interconnections. The gun will come last with all the optical elements supplies.

Figure 4: the buncher and two magnetic lenses on their girder, in the tunnel

LT1 transfer line

16 meters long, LT1 consists mainly of magnets, vacuum chambers and diagnostics.
Besides transferring the beam from the LINAC to the booster, it is essential for testing the LINAC performances. It is now located in the tunnel.
The magnets have been measured. Their power supplies (LT1 power cubicle), tested at the SOLEIL laboratory for three months, and the vacuum chambers, tested at LURE vacuum laboratory, met the expectations.
The diagnostics include position screens (delivered, aligned and adjusted at SOLEIL), current and charge monitors (delivered and tested several months ago), an energy analysis slit (already tested, soon to be delivered) and the emittance monitor (already validated and currently under test at the diagnostics laboratory).
The control-command system consists of a fully tested and validated hardware architecture, and, on the software side, of a great number of Tango Devices servers controlling each sub-part of the equipment. They have been developped as generically as possible (i.e. with the view to be reused in other cases than the LT1) and are definitively tested as soon as some equipment or sub-part of equipment is delivered to SOLEIL. In addition, thanks to the software gateway written by the ICA[1] group between TANGO and the scientific MATLAB software, the Machine Physics group has validated the control of some equipments using the MATLAB software, such as for example the emittance measurement (screen, camera, optical system, video processing). The remarkable ease with which physicists, non-specialized in computer sciences, are able to manage the control of their equipment has been confirmed .

The installation of the Booster is impending

All the magnets (37 dipoles, 45 quadrupoles, 28 sextupoles and 44 correctors) have already been delivered. The measurement campaign is now completed for the dipoles, the quadrupoles and the sextupoles. A first girder is ready (figure 5) for installation in the stride of the LINAC and LT1. A number of chambers are already stored at LURE where they are being tested. Girders and frames are delivered. The accelerating cavity (coming from the LEP) has already been equipped and tested. It is ready to be installed in the tunnel as well as its 35 kW emitter (Re. Rayon de SOLEIL #10).

Figure 5: The girder, with its magnetic elements and vacuum chamber

The injection and ejection pulsed elements of the beam as well as their power supplies are expected by the end of the year. The power supplies for the correctors and sextupoles are ready. The 3 Hz dipole and quadrupole power supplies will come last, in the course of February 2005 (a power supply prototype will be delivered shortly). The Commissioning[2] of the Booster will then be able to start.

[1] ICA : Data Processing Control and Acquisition; *re:* page 18.

[2] commissioning: life size test phase, to last until the expected performance is achieved

Magnetic measurement campaigns: an essential stage

The booster, the storage ring and the LT1 and LT2 transfer lines (from the booster to the ring) consist of almost 400 magnets of 3 different types: the dipoles (bending of the trajectory), the quadrupoles (focusing) and the sextupoles (compensation of the non-linear effects). A good quality of these magnetic fields is essential to ensure the announced performance of the electron beam. Tolerances have therefore been defined for all types in terms of reproducibility, identity from one element to another, transverse homogeneity. Magnetic measurements are therefore an essential stage since are used to validate the prototype magnets before launching the manufacturing of the series and subsequently to validate and characterize all the magnets of the series. Several measurement benches have been designed and built in order to optimize the precision and reproducibility of the measurements for each type of magnet (see results in the box below)
Finally, all the data deduced from the magnetic measurements will serve to improve the model used for the beam dynamics and thus to prepare the most realistic operating starting point.

About the storage ring

Measurements and studies are being conducted in parallel with the preparation of the installation

A prototype girder

A long prototype girder (4.80 m) was built to study the static aspects (deflection, flatness) and dynamic aspects (stability), to ensure that the specifications are fulfilled and to validate the models computed with a 3D (ANSYS) code. These measurements enabled us to bring several improvements to the design and also to validate the computer code.

Figure 6: Test bench. The magnetic elements are replaced by loads of identical weight

The girder supports 4 quadrupoles and 3 sextupoles. The dipoles are installed at each end of two adjacent girders. For the test bench, dummies of identical weight and equivalent moments of inertia simulated the magnetic elements and the adjacent girders have been replaced by 2 blocks of concrete (figure 6).

Example of results: magnetic measurements of the 38 dipoles of the storage ring type

We are dealing here with the 32 dipoles for the storage ring + 4 for LT2 + 1 reference + 1 spare. Two types of benches are used: the BCH15 bench for establishing the mapping (s, x) in the median plane with a 15 probe Hall package and the BCD bench (figure A) for measuring the difference in field integral on the axis with respect to a reference dipole through the stretched wire method.
The mapping with the BCH15 bench is used to obtain the field integral on both the magnetic axis and length, the difference in magnetic length between entry and exit, which may influence the final positionning in the ring, and finally to reconstitute the actual curved trajectory taking into account the fringe field. The transverse homogeneity around this trajectory, the position of the original points of the radiation, and a possible shift of the trajectory at the exit of the dipoles with respect to the reference orbit can then be deduced. The measurements of 4 dipoles have shown that the magnetic length (1055.5 mm) is identical from one dipole to another by some 10^{-4}. It is slightly larger than expected (1052.43 mm) and we will reduce the nominal field from 1.7114 T to 1.7065 T for the storage ring to operate at 2.75 GeV. The analysis of the transverse homogeneity leads to multipolar components close to those worked out at the design stage, i.e. better than the tolerances ($\Delta BL/BL$ total < 5.10^{-4} to x = ± 20 mm). The tolerance specified for the identity of the dipoles is 10^{-3} (in RMS deviation) and measurements lead to a value of 6.10^{-4}, which is satisfactory.
In order to reduce the horizontal natural orbit distortion in the storage ring, a sorting program has been developped and applied to the 32 dipoles. It allowed us to determine the optimal position of each dipole in the storage ring.

Figure A: BCD test bench

First of all, some improvements have been made, such as the modification of the position of the jacks to reduce the static stress under load, and the suppression of the central stand.
The dynamic study was entrusted with the company AVLS which had already worked for SOLEIL for the on-site vibrations study: a program including measurements of the ground response, a modal analysis of the girder with and without load, as well as measurements of responses to various excitations has been achieved.
The first mode of the girder under load is 44 Hz (specification: first mode higher than 40 Hz) which is in very good agreement with the computation. The effectiveness of the blocking of the jacks for the reduction of vibrations has also been verified. The measurements brought to light a natural dipole frequency mode at 14 Hz (value which was subsequently corroborated by computation). This low frequency mode was liable to have an adverse impact on the stability of the beam, although this stability depends on the quadrupoles linked to the girder and not on the dipoles. A new more rigid dipole linking system on the girder has been studied, which however ensures some amount of flexibility necessary to sustain thermal expansions. Finally, a new series of measurements confirmed the increase of the first dipole natural frequency at 27 Hz and that of the girder at 46 Hz giving us very good confidence in the quality of the girders for the stability of the beam.

Vacuum chamber impedance

The performances expected for the electron beam produced by SOLEIL may be limited by the interaction between the beam and the vacuum chamber (see box). The knowledge and the minimization of the coupling impedance between the beam and its vacuum chamber becomes therefore of great significance for high intensity multibunch and temporal structure operation modes.
The various elements of the machine have been studied with the 3-D Gdfidl code implanted on the processor cluster developed by SOLEIL. This enabled us to use an extremely fine meshing (about a tenth of a mm) and to integrate the wake field over several meters (6 m) within reasonable computing time. Indeed, this fine mesh size is necessary considering the size of the bunch (6 mm), and the integration distance must be sufficient to ensure a good convergence of the solutions, in particular for the computation of the modes trapped in cavity like structures.
This study brought about, on the one hand, the modification of some elements of the vacuum chamber, and, on the other hand, the evaluation of the global impedance seen by the beam.

Different types of beam instabilities

There are two main types of beam instabilities due to the interaction between the beam and the vacuum chamber:

- in multibunch, or "high brilliancy" mode (large number of electron bunches very close to one another,) occur longitudinal and/or transverse oscillations of bunches coupled together. This type of instability is caused by the narrow band impedances of the trapped modes driven by cavity like structures.
- in single bunch, or "temporal structure mode" (small number of bunches of very high intensity), the instability shows a bunch lengthening associated with a widening of the dispersion in energy, and, in the transverse plan, a limitation of the current per bunch due to the coupling of the transverse modes (TMCI). This type of instability is caused by the broad band impedance of the vacuum chamber (Re: Rayon de SOLEIL # 8).

- the dipole chamber

Available data in the literature indicated that the opening of the slit separating the vacuum chamber and the anti-chamber of the dipole should be of 9 mm. Studies on different insertion devices have shown that this value was not easily compatible with the users' needs in vertical polarized radiation and a vessel design allowing sufficient cooling. Following the computation, the opening could be increased to 15 mm with no modification to the impedance which is dominated by other components. This is therefore a good news for the vacuum chamber design of the synchrotron sources.

- flanges

The initial design had a 0.4 mm thick, 50 mm deep slit. The calculation revealed a large amount of trapped modes of relatively low frequencies and high impedance which would have led to unacceptable multibunch instability thresholds. A short circuit was introduced between both walls, as shown in figure 7, which turns out to be very effective for shielding the structure.

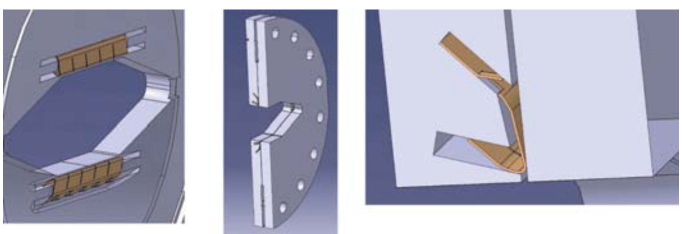


Figure 7: New design of the flanges with the HF short circuit. The zoom shows the residual cavity which has come down to 0.4 mm x 1 mm

- *Beam Position Monitors (BPMs)*

The calculations have shown that in multibunch mode, as well as in temporal structure mode, a power higher than 1 W was deposited on each electrode. This power is very difficult to evacuate and the thermal study has led to the estimation of an excessive temperature, liable to induce differential expansions and consequently, a dangerous stress of the ceramic as well as measurement pollution. Increasing the thickness of the electrode from 2 to 5 mm allowed us, while keeping the same measurement resolution, to divide by 2 the deposited power, which becomes totally acceptable.

Effect of the insertions on beam lifetime in SOLEIL

When an undulator (or a wiggler), even a perfect one, is inserted, in a straight section of the machine, the 4^{th} order symmetry is broken. The focusing effect of the undulators leads to a beat of the optical functions which must be compensated for. Moreover, its magnetic field may excite resonances which affect the dynamic acceptance and, therefore, the lifetime of the beam. SOLEIL will comprise a large amount of undulators, the great majority of which consists of helical undulators to ensure variable polarization, and low gap in-vacuum undulators (~ 5 mm). Owing

Global impedance results

The effective impedances (i.e. impedances convoluted with the beam spectrum) have been worked out in the multibunch and temporal structure modes for most components. Longitudinally we are used to characterizing the impedance quality of a machine by the absolute value of the effective impedance divided by ω/ω_0 where ω_0 is the revolution pulsation. In the case of SOLEIL, $|Z/n| = 0.2\ \Omega$ half of which coming from the resistivity of the wall. The small contribution of the other elements is fully satisfactory and reflects the particular care brought to the design of the different vacuum chambers in order to soften any discontinuity.
In the transverse plane, it is the product of the effective impedance by the local beta function which is the relevant quantity for the collective effects. In the vertical plane, we obtain for the real part and the imaginary part ($\sum\beta_v$ x (Z_v)eff) : 0.3 MΩ and 1.4 MΩ, respectively. On the horizontal plane, the real part is divided by 3 and the imaginary part by 2.
Figure B shows the dependency in frequency of the longitudinal impedance for the most important elements. Impedance due to the wall resistivity has been assessed in the most precise way by using the characteristics of each element (transverse section, length, thickness of the wall, electrical resistivity, NEG coating, asymmetry of the chamber, beta function, etc.). Results confirm that the wall resistivity contributes by half to the total impedance and, furthermore, that the imaginary part is increased by a factor of almost 2 by the NEG coating which has a particularly important effect in the insertion devices chambers. The other main contributions come from the bellows, the BPMs, the RF and the insertion chamber transitions. For the three planes (longitudinal, horizontal and vertical) the impedance is mostly inductive but shows a resonance of about 10 GHz coming from the BPMs.
Some elements remain to be evaluated but their contribution should not change significantly the results and the collective effects will be worked out on this basis. The instability thresholds of the "resistive-wall", micro-wave, transverse coupled modes etc. will be precised as they have been previously assessed with a rather large degree of uncertainty. We will also be able to assess more accurately the length of the bunches above the micro-wave threshold, according to the intensity of the bunches. This value is particularly important for users of the temporal structure mode.

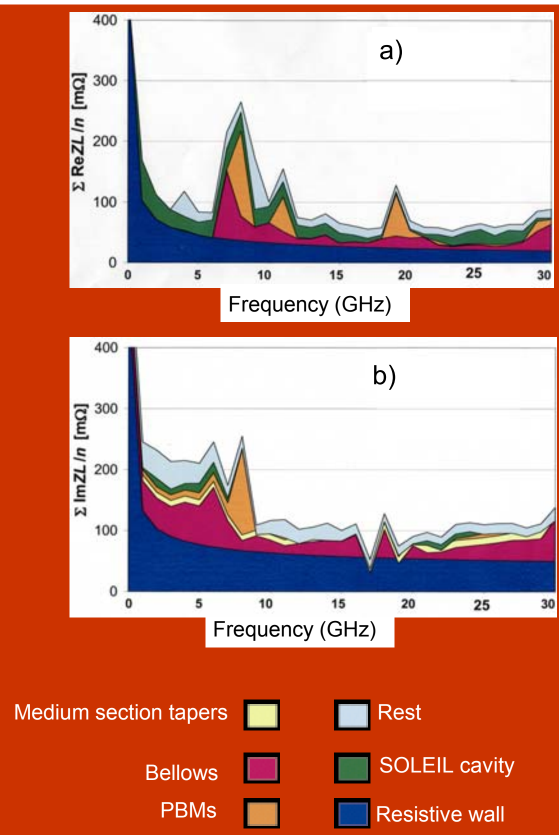


Figure B : Real (a) and imaginary (b) parts of the longitudinal impedance according to the frequency for the main vacuum chamber components.

to the complexity of these magnetic fields the model for standard fields becomes inapplicable. Therefore the RADIA 3D [1] code was used to generate maps describing the field profile in both horizontal and vertical planes. The optics codes BETA and TRACYII were then modified in order to track the particles in presence of these field maps. In addition, TRACYII has been associated to a Frequency Map Analysis program [2] to scrutinize more finely the dynamic acceptance and identify the most dangerous resonances. Studies have been carried out on the initial adjustment for phase 1 undulators configuration (three U20 in-vacuum undulators and the helical undulators, HU640, HU256 and HU80). Their major characteristics are given in the table below.

Undulator	U20	HU640	HU256	HU80
Length (m)	1.80	8.96	3.328	1.52
Period (mm)	20	640	256	80
Max field V (T)	1.02	0.11	0.400	0.925
Max field H (T)	-	0.09	0.275	0.718

Table 1 : Characteristics of the phase I undulators

For the HU640 and HU256 undulators, the focusing effects are negligible and the effect on dynamic acceptance is not significant.
The vertical focusing introduced by U20 is weak, but the transverse homogeneity of the vertical field (Figure 8) is insufficient to accept particles with an energy tune shift beyond – 4.4%. This limitation can only be eliminated with unrealistic pole widths (100 mm).

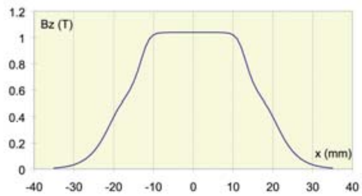


Figure 8: Transverse homogeneity of U20 field

The effect of HU80 undulator has been studied for the three polarization modes of operation:
Linear horizontal, linear vertical and helical modes. In any cases, the inhomogeneity of the fields led to an important focalisation in both planes and the compensation of the optical functions beat is necessary for the linear vertical mode.
Besides, the Frequency Map Analysis has shown that the restoration of the symmetry cancelled the drastic effect of a dangerous resonance.
The working point has been optimised again, taking into account the cumulated effect of all phase I insertion devices.
The new nominal point (ν_x = 18.19; ν_z = 10.29) leads to:

- A good “on-momentum” acceptance (fig. 9) necessary to ensure a very good injection (even in top-up operation) and a sufficient beam lifetime due to the Coulombian collisions on the residual gas.
- A good “off-momentum” acceptance necessary to ensure a sufficient Touschek beam lifetime

The Touschek beam lifetime (calculated with a 6D tracking) with all phase I insertion devices is almost 27h. Combined with a beam lifetime due to residual vacuum of almost 24h, it leads to a total beam lifetime of almost 15h.

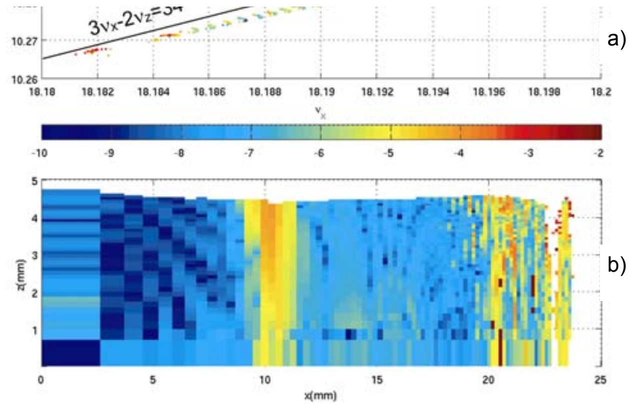


Figure 9: (a) Frequency map describing the variation of the wave numbers with horizontal and vertical oscillation amplitudes. (b) “On-momentum” dynamic acceptance of the new nominal point. The diffusion rate of the wave numbers is coded with colours according to a logarithmic scale, from 10^{-10} (blue) to 10^{-2} (red) which corresponds to a variation of the wave numbers respectively of 10^{-10} and 10^{-2} after 1000 turns. The blue colour corresponds to areas where particles motion is very stable, red corresponds to areas where particles motion is non-linear, even chaotic and white colour means that the motion is unstable. Each couple of wave numbers (a) corresponds to a couple of amplitudes x, z (b). The visible resonance $3\nu_x$-$2\nu_z$=34 on the frequency map expresses at x=10 mm in dynamic acceptance. Its effect is not dangerous because it is bounded by two horizontally stable areas (blue area) and its vertical diffusion (orange area) corresponds to a rate of 10^{-4}.

Contact: Marie-Paule Level
marie-paule.level@synchrotron-soleil.fr

[1] O.Chubar, P. Elleaume and J. Chavanne, “A 3D Magnetostatics Computer Code for Insertion Devices“, Journal of Synchrotron Radiation, 1998, vol.5, p.481.

[2] J. Laskar, Frequency Map Analysis and Quasiperiodic decompositions, proceedings of the school “Hamiltonian systems and Fourier analysis“, Porquerolles, sept. 2001.

Hard X-rays for structural studies

This is brief presentation of the characteristics of the 3 beamlines, DiffAbs, ODE and PROXIMA 1, and their state of progress. These beamlines, using the SOLEIL X-rays, will be used for the structural characterization of samples in numerous disciplines: biology, physics-chemistry of materials, magnetism.
These beamlines, part of the first group of lines installed at SOLEIL (phase I beamines), will be open to the scientific community in 2006.

DiffAbs: structural characterization of materials

The DiffAbs beamline will be constructed by the partial transfer of H10; the last beamline built at the DCI ring of LURE. At SOLEIL, DiffAbs will be dedicated to the structural study of a great variety of materials combining, if required, X-ray diffraction, X-ray fluorescence and X-ray absorption spectroscopy. This beamline, in particular, is part of the program of collaboration developed between SOLEIL and the Région Centre.

For instance, the DiffAbs beamline will offer the almost simultaneous combination of X-ray absorption spectroscopy (EXAFS[1]) and wide angle scattering (WAXS[2]) in order to obtain information about the short and medium-range characteristics of the studied materials. Furthermore, resonant diffraction spectroscopy, such as DAFS[3], will be used to get information with site and chemical selectivity. More generally, diffraction/scattering will be used with anomalous effect (variation of the scattering cross section in the vicinity of an absorption edge of the probed element). Finally, time-resolved X-ray diffraction measurements (millisecond measurement time) will be available to study fast kinetics.

A bending magnet beamline

Two bent, Rh-coated mirrors and a double-crystal Si (111) monochromator will provide a monochromatic focused beam in the 3-23 keV energy range ("normal mode"). By utilizing an additional, secondary focusing optics, e.g. a Kirkpatrick-Baez (KB) mirror-pair, a microbeam of several microns size will be available ("microbeam mode"). The characteristics of these two operation modes are summarized in table 1. In the microbeam mode smaller spot-size with reduced flux can also be achieved.

	Utilisation modes of the monochromatic beam	
Parameters	**"normal"**	**"microbeam"**
Spectral range (keV)	3 – 23	3 - 19
Energy resolution	~ 10^{-4}	~ 10^{-4}
Beam size (HxV, FWHM in mm^2)	0,140 x 0,240	0,014 x 0,011
Flux ($ph.s^{-1}$)	10^{12} - 10^{13}	10^{10} - 10^{11}
Divergence (mrad)	variable H = 7.14 – 2.90 V = 0.54 – 0.20	variable H = 2.87 – 0.98 V = 6.22 – 2.14

Table 1: General characteristics of the different beam modes available at DiffAbs.

Beamline layout

The DiffAbs beamline will consist of two safety hutches, a control room, a seminarium/data-processing room and a workshop at the end of the

APPLICATIONS: HIGH TEMPERATURES

An important scientific activity of DiffAbs will be the study of the behaviour of materials at very high temperature (T < 3000°C). Thanks to the dedicated experimental techniques, the molten and overcooled states (metastable state, when the material is liquid below the melting point) of oxides, ceramics and metal alloys can be studied. Different crystalline phase transitions or the behaviour of certain powders during their transitions into dense and nano-structured materials (self-propagating high-temperature synthesis) can also be investigated.

[1] EXAFS, Extended X-ray Absorption Fine Structure
[2] WAXS, Wide Angle X-ray Scattering
[3] DAFS, Diffraction Anomalous Fine Structure

The tender procedure of the diffractometer is in the final state, the study and construction phases will begin at the end of 2004. The installation of the goniometer at DiffAbs is scheduled for the end of 2005.

beamline. The safety hutches will provide the necessary shielding against the scattered and direct radiation. The thickness of the sandwich structure (Pb between two steel layers) of the walls of the hutches ensures that the total dose rate does not exceed the valid norm. 0.5 μSv.h^{-1}. The optics hutch will host all the primary optical elements providing monochromatic focused beam in "normal" operation mode. The secondary focusing optics, the diffractometer and the experimental environments will be situated in the experimental hutch, equipped with somewhat thinner shielding walls. The tender procedure of the hutches is in the final phase; their detailed study and manufacturing will start during the second semester of 2004, while their mounting at SOLEIL is scheduled for the spring of 2005.

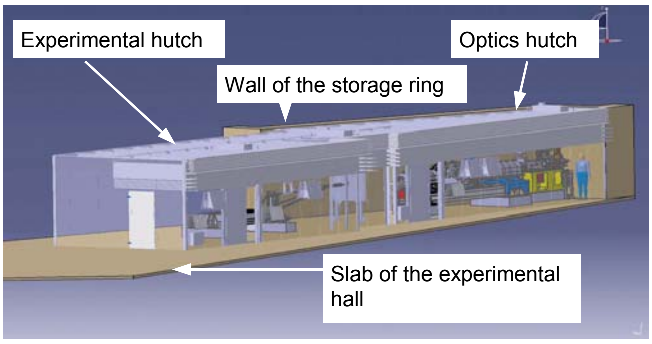


Figure 1: 3D image of the two safety cabins of the DiffAbs beamline. The walls are transparent in the figure in order to show the optical elements, which have already been studied and optimized.

DiffAbs will be equipped with a 6 circle diffractometer foreseen to be used for most of the experiments. This high precision mechanical instrument will consist of two concentric goniometers; a 4 circle goniometer ensures the orientation of the sample, while the detector can be oriented by 2 additional circles. The geometry retained for the orientation of the sample is a kappa geometry, in which the three main circles are a combination of the Euler angles. This geometry is used to correct and improve the mechanical performance of the goniometer. It also has the advantage of ensuring a large free volume for the installation of heavy and voluminous samples and their environment.

OTHER APPLICATIONS…

The study of thin films and interfaces, in particular to determine their mechanical properties (strain measurements and stress determination), will be another important application field of DiffAbs. It will also be possible to perform micro-diffraction, micro-XRF and micro-XAS experiments at the beamline. These analytical possibilities will provide useful research tools in the fields of material science, environmental science as well as in the study of samples of cultural heritage.

Available sample environments

A furnace developed for the H10 beamline will be transferred and will be available at the DiffAbs beamline. This furnace, designed to study thin films or powders, operates in air at temperatures up to 500°C and in vacuum up to 800°C. The purchasing of a new furnace working up to 2000°C in vacuum or in controlled atmosphere is foreseen in collaboration with the CRMHT[4] (Orléans). Furthermore, the device applied for the study of oxides in molten state at H10 will be upgraded by including two CO_2 power lasers for heating and a chamber for the aerodynamic levitation of the sample (see figure 2).

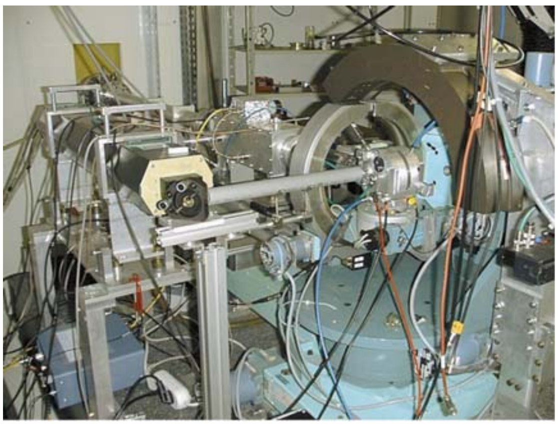


Figure 2: Experimental device at H10 for the study of high temperature oxides. It includes a CO_2 laser, a sample chamber located on the diffractometer for aerodynamic levitation and EXAFS measurements, and a curved gas detector (INEL 120°) for recording diffraction spectra.

[4] CRMHT, Centre de Recherche sur les Matériaux à Haute Température

In the framework of a French-German collaboration (CRMHT, SOLEIL and the DLR[5] in Cologne), another device is under development for the study of metal alloys in molten and undercooled states. Aerodynamic levitation will be used to avoid the contact of the liquid sample with the sample holder. The heating of the sample will be achieved by a radio-frequency generator. Preliminary feasibility tests have been carried out with this setup at the D2AM beamline of the ESRF[6]. Finally, in collaboration with several laboratories, it has been envisaged to develop an in situ tensile tester with a heating system for studying the thermo-mechanical properties of certain materials.

Contact: Dominique Thiaudière
dominique.thiaudiere@synchrotron-soleil.fr

[5] DLR, Institut für Raumsimulation, Deutches zentrum für Luft-und Raumfahrt
[6] ESRF, European Synchrotron Radiation Facility

ODE: structural characterization of materials under extreme conditions and under kinetics

The ODE beamline corresponds to the partial transfer of D11, a LURE beamline upgraded as early as 1996 and adapted to be transferred on a third generation storage ring.

The ODE scattering EXAFS spectrometer is dedicated to X-ray absorption measurements: XANES, EXAFS and XMCD[1]. Its particularity is to select at once all the energies of an absorption spectrum by thanks to a curved Bragg reflector (figure 1). This assembly has several advantages, the major one being that the beam can be focused on the sample in a completely static way. There is no mechanical movement of the lens whatsoever during the acquisition of a spectrum. This stability made it possible to carry out very accurate measurements on the DCI first generation ring at LURE where absorption variations in the order of 10^{-4} (XMCD at the threshold K of 3d metals) were detected. The size of the focusing spot will be about 40 µm at SOLEIL, making it possible to work on small samples such as those used under very high pressure.

In addition, high speed acquisition can also be achieved with this assembly. Here also the absence of mechanical movement makes it possible to record in one go, and in a very short time, an X-ray absorption spectrum. The shortest time expected at SOLEIL for an EXAFS spectrum with a good signal/sound ratio is 1 ms.

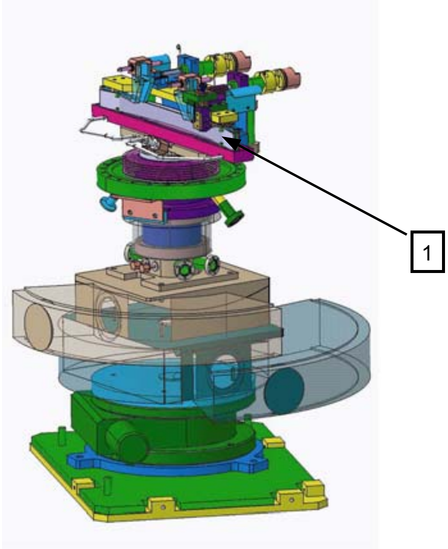


Figure 1 : monochromator transferred from LURE, adapted from an original design developed for the ESRF ID24 line

A curved blade of silicon (1) selects at once all the energies of an absorption spectrum and focuses the beam on the sample

At LURE, one of the limits of this spectrometer was the domain of energy selected for energies below 7 KeV, giving a limited EXAFS domain. With wider lenses than the previous ones it will be possible to push back this limit.

Work in transmission and fluorescence

The work mode in scattering is transmission. However, the work mode in fluorescence will become possible thanks to the shift of a slit in the beam. This will result in an acquisition time comparable to that of the conventional EXAFS stations but will leave the focusing point completely independent from any mechanical movement and, therefore, very stable.

[1] XMCD : X Ray Magnetic Circular Dichroism

Broached themes

The ODE spectrometer is used for the magnetic circular dichroism measurements of hard X-rays (XMCD) and EXAFS measurements under very high pressure. The small dimension of the beam at the level of the sample allows us to envisage pressures in excess of one Mbar. The recording in a single reading of the absorption spectra detector allows us to visualize in real time the transmission of high pressure cells and therefore optimal adjustment of their position and their orientation in order to obtain a measurement free of parasite distortion and reflection induced by the diamonds of the cells.

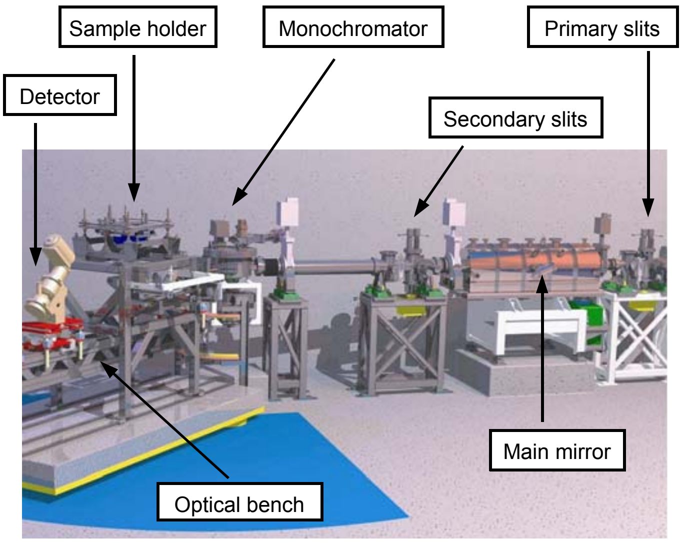


Figure 2 : the spectrometer

APPLICATIONS

Besides the XMCD experiments and EXAFS measurements under high pressure, the possibilities of study offered by ODE in other fields are many and little exploited up to now. The X-ray lens of the spectrometer in scattering mode will provide a 40 µm probe beam able to penetrate the heart of reactional enclosures with transparent windows for hard X-rays (mylar, capton, mica, diamond, beryllium, sapphire etc.). This gives the possibility of making in-situ measurements of small samples of interest to chemistry, electrochemistry, biology, catalysis, geochemistry and environmental sciences, as well as corrosion, etc.

Highlights of the progress of the line

ODE will consist of two radioprotection cabins, a control room, a room dedicated to the users, and a workshop at the end of the line. The cabins will be armored with steel and sandwiched between two steel plates so that the total dose rate does not exceed the valid norm of 0.5 $\mu Sv.h^{-1}$. Invitation to tender for the construction of these cabins is out. The first one, or "optics hutch", will comprise all the optical elements, from the exit at the front end of the line to the monochromator (figure 2). The experimental hutch will contain the optical arm supporting the sample environment and the detector.
The characteristics of the lenses are under definition, the main mirror is under construction. Orders for the other elements in the optical cabin will be placed in the last quarter of 2004. The designs of the optical arms and the sample environment have been finalized in September 2004.
A new CCD camera has been ordered, the software is under development, particular care is given to the control and processing of information read on the detector

Contact: François Baudelet
francois.baudelet@synchrotron-soleil.fr

PROXIMA 1:
First stop at the "Protein village"

Scheduled to be operational by January 2006, Proxima 1 is a beamline on an undulator source, the U20.The X-rays produced will be of high intensity, low divergence and very stable: all essential beam attributes for future measurements in bio-crystallography.

What field of study?

All living creatures consist of cells – small functional units surrounded by a cell wall. The most simple organisms consist of a single cell, and the most complex (plants, animals, man, etc.) consist of billions of cells, each playing its own part in the functioning of the organism (conversion of light into energy, digestion, etc.). A very complex internal chemistry ensures the survival and the function of these cells. This so-called biochemistry uses large molecules (nucleic acids, proteins), supported by smaller ones (vitamins, minerals). In spite of significant progress in biology, the comprehension of the functioning of living beings at the molecular scale is one of the great challenges of science in the 21st century.

Microscopes enable us to observe a cell, but not to see the functioning of proteins and nucleic acids, nor the interactions which link them.

An in-depth understanding of the functioning of living beings would enable us to cure diseases or to improve the quality of life. Here is an example : Valérie Biou (researcher at the enzymology laboratory in Gif sur Yvette) has been working on the proteins of plants (figure 1). Plants "don't eat", and therefore must produce themselves the amino-acid "building blocks" necessary to produce those proteins that they can't draw from the soil. These amino-acid production mechanisms are specific to plants since animals derive the necessary amino-acids from their food. If we could find out the structure of the proteins of plants which produce amino-acids and understand their biochemistry, we could try to specifically block the production of certain amino-acids. When plants become unable to produce these amino-acids they die, destroyed in a selective way (not a problem for animals). Herbicides bought in garden centers use a mechanism of this type. In the same way, the study of certain bacterial proteins allows us to understand the mechanisms of resistance to antibiotics developed by these bacteria, and to produce new and more effective medicines.

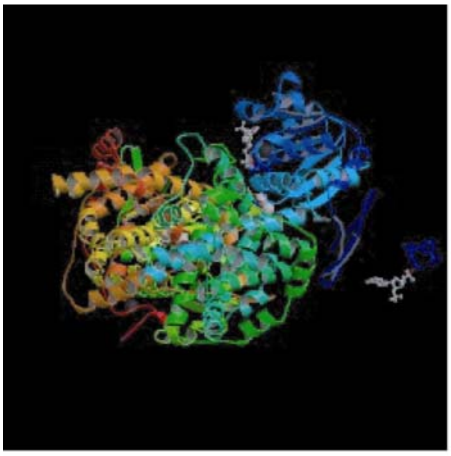


Figure 1: A ribbon diagram representation of the folding of the amino acids making up the enzyme acetohydroxyacid isomeroreductase from spinach. The structure of this protein, part of the enzymatic reaction for synthesising the amino-acids valine and isoleucine, contains 17729 non-hydrogen atoms.

V. Biou et al., *EMBO J.* **16** (1997), 3405.

BIOCRISTALLOGRAPHY

This is the most widely used method for studying the 3D structure of large biological molecules or macro-molecules at high resolution. The technique consists in producing a large amount of the studied macromolecule, purifying and crystallizing it. Each step may take years, in particular for "difficult" proteins, or complexes of several proteins. Diffraction images are then recorded, taken by "hitting" the crystals with X-rays produced by a synchrotron for instance (see figure 2).

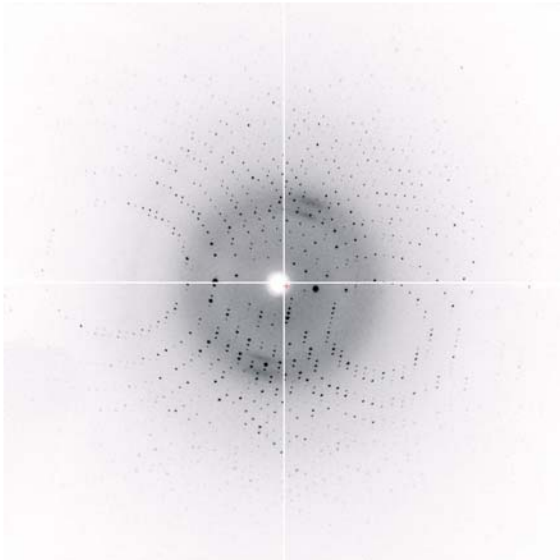
One limiting stage of structure determination is the "phase problem", the link between the measured diffraction data and the calculated 3-dimensional structure. Once this problem solved, one moves to interpretation of the structure: "rebuilding" the studied molecule by mapping out the electronic density, imposing chemical constraints and understanding how the molecule fulfils its function.

Figure 2: A goniostat supporting a crystal sample

The movement of the rotation axes of the goniostat orients the crystal with respect to the beam (which enters on the left). The detector (not shown here) is located on the right.

Figure 3 : Image de diffraction de la protéine trypsine, acquise à l'ESRF

Chaque point noir est une réflexion, la symétrie de la diffraction est mise en évidence.



Analyze thousands of diffraction spots to rebuild the 3D structure of the molecule

A diffraction pattern must be recorded for each crystal "3D orientation", in order to examine the diffraction of the crystal in all possible directions. The image of a typical diffraction is shown in figure 3. The "symmetrical" organization of the diffraction spots (reflections) indicates the symmetry of the molecules in the crystal (cubic for instance). The intensity of the reflections contains more subtle and useful information: the relative positions of all atoms in the molecule. The positions and intensities of the reflections are measured (there can be tens of thousands per diffraction pattern!) with great precision, and mathematical methods are then used to reconstruct an image of the molecule. Experiments are often repeated with modified molecules (for instance in the presence of drugs, vitamins or small molecules which influence the functioning of the protein) for a better understanding of its function.

Biocristallography at SOLEIL

The PROXIMA 1 line, under construction, will provide a focussed beam for the study of small crystals (often smaller than 50 microns, sometimes as small as a few microns). The tuneable wavelength will make it possible to use mathematical processes of reconstruction of the molecule by using the MAD[1] method (an experimental protocol which facilitates these mathematical processes by using measurements of the diffraction with X-rays of different energies. The necessary control of the X-ray energy is only possible on a synchrotron).

The main beamline optics (focussing mirrors and a monochromator) have been ordered, and the equipment for orienting and positioning the crystal samples is currently being chosen

The main asset of SOLEIL for future biocrystallography measurements is the stability of the source which will make it possible to study smaller crystals: **the proteins which are most difficult to crystallize** (and therefore giving small crystals) often being scientifically the most interesting!
The biological revolution marches on...

[1] MAD, Multiwavelength anomalous diffraction

Contact: Andrew Thompson
andrew.thompson@synchrotron-soleil.fr

X – ray Absorption Spectroscopy for materials science: from LURE to SOLEIL

This thematic CNRS training course held from 6 to 12 June 2004 at Aussois, was meant for "regular" users of the LURE beamlines and future partners of the scientific activity developed at SOLEIL. The fields of research covered by this project were the materials science, the surfaces, environmental and chemical sciences.

Financed for the greater part by the CNRS and managed by the Paris B regional delegation, this training course also received financial support from the LURE and SOLEIL laboratories. The training was attended by 58 persons (including 18 training contributors): half of them CNRS agents, university professors, CEA agents, SOLEIL or ESRF researchers and doctoral students from 33 laboratories. More than half of the participants were chemists, which shows that X-ray absorption spectroscopy has become a full-fledged characterization technique in chemical laboratories and that this community intends to carry on this activity on SOLEIL. We haven't mentioned life sciences disciplines which are dealt with during specific courses: the Bio-XAS symposiums.

A threefold objective

The primary aim of this course was to deal with the evolutions of the theoretical approaches to data analysis required by the move to a 3rd synchrotron generation.
Besides, the presentations enabled beamlines first scientists to talk about the future SOLEIL absorption beamlines and research paths that could be developed there in collaboration with the users.
Finally, it was an opportunity to keep up with the links developed at the LURE between members of the "X-ray absorption spectroscopy" community during the "dark period", the transition period between the LURE shutdown and the beginning of operation of SOLEIL. The aim of this gathering between the persons in charge of absorption beamlines and its future users was to define together the scientific activity that will be developed at SOLEIL.

The training started with a detailed presentation of the processes involved in X-ray – matter interaction and more specifically in absorption phenomena. Then came an exhaustive presentation of the most recent mono and multi-electronic calculation codes, some still under development, and which cover the entire energy range available at SOLEIL. The most recent works on new methods for calculating absorption spectra were also broached. The courses, completed with two round tables, showed the interest of coupling X-ray absorption with molecular dynamics or other spectroscopic techniques such as photo-emission. According to the participants, the presentations were of a very high level, both clear and pedagogic.
After taking stock of the progress made by the SOLEIL project and recalling the characteristics of the photons delivered by this new machine, exhaustive presentations were made dealing with the future SOLEIL absorption beamlines: LUCIA, SAMBA, ODE, TEMPO, DiffAbs.
The scientists in charge of these beamlines made our mouth water by setting out the prospects open by their new instruments. Users were able to express their needs during workshops and round tables on the research themes developed by the project designers of the new SOLEIL lines. The main themes broached were dichroism, the coupling of techniques, micro-spectroscopy and studies carried out under extreme conditions.

A public asking for more!

During the end of the week assessment, the participants said they had found this course very interesting and useful and underlined the importance of the contacts established with the persons in charge of the future SOLEIL beamlines.
The objective which was to help transfer of the X-ray absorption community from LURE to SOLEIL has thus been achieved. It is important to maintain the high scientific level of this type of course and to keep the community informed of the developments in fundamental techniques and theories. The participants considered that such a course was essential before the start-up of SOLEIL scheduled in June-July 2006.

The content of the courses and presentations are available since September 2004 in the form of a CD.

An environment favorable to thinking...

Encouraged by the commentaries and the participants' request, we intend to repeat this course when the SOLEIL beamlines are open to users (summer 2006) by integrating into the organization committee the persons most liable to fulfil the expectations formulated during the closing round table.

Contact: Christophe Cartier dit Moulin
cartier@ccr.jussieu.fr

Light on Heritage

On the occasion of the European Heritage Days dealing this year with "Heritage, sciences and techniques", synchrotron SOLEIL organized an event entitled "Light on Heritage" within the Research and Restoration center of the French museums (C2RMF, Paris).
On the 18th and 19th September, a mixed public of about 1000 persons was able to discover the possibilities offered by synchrotron radiation for analyzing heritage materials. Applications were illustrated by concrete work carried out by the C2RMF teams, from the study of Dürer drawings using X-ray micro-imaging to the aging of biological tissues through the observation of micro-diffraction diagrams of mummies hair. The exhibition included a slide show.
This initiative, reported by the media (Libération, RFI, France Inter, Europe 1), received also a very positive echo from the institutional partners of SOLEIL who visited the show (representatives of the Minister of Culture, Ile-de-France region, CAPS, French museums etc. including the Ministry of Research himself). This was the official launching of the archaeology & cultural heritage interface of the synchrotron SOLEIL.
The show should now travel in the framework of the 2005 Worldwide Year of Physics, in Ile-de-France, Brittany etc.

Contact: Loïc Bertrand
loic.bertrand@synchrotron-soleil.fr

Workshop "Micro-diffraction: mechanics of materials and phase identification"

Following the "Imaging and synchrotron SOLEIL" workshop of November 2003 which had highlighted a strong demand for micro-diffraction, the goal of this workshop was to determine the analytical needs of the various scientific communities involved in this field and to initiate a thinking tank at the technical level in terms of beamline and/or specific equipment.

Organized on the 28-29 June at the Curie Institute of Orsay on the initiative of SOLEIL and the French Crystallography Association, this workshop also benefited from the joint financing of the Federation of mechanics and materials, structures and processes of the Ile-de-France region (*Fédération francilienne de mécanique des matériaux, structures et procédés).* More than 90 persons attended: users of the LURE, the ESRF and a large number of potential users. The spontaneous attendance of half a dozen industrialists and representatives of several major specialized research organizations (INRA, CEA/LETI, CEA/DEM) should be noted, as well as the presence of about ten doctoral students, recipient of a scholarship awarded by the French Crystallography Association.
The workshop was held in three sessions focusing on the constraints and mechanics of materials, the identification and analysis of phases and instrumental aspects. The presentations of the 17 speakers, among whom three from Germany and one from the United States, underlined the diversity of the fields of research covered by this technique: microelectronics, metallurgy, earth and environmental sciences, Heritage, bio-materials, plastic and chemistry. Presentations by colleagues from other synchrotron centres showed that interest in micro-diffraction is rapidly growing abroad.
During the round table that closed the workshop, it was decided to set up a certain amount of work groups, defined according to scientific themes and technical needs, with the view to prepare the necessary elements for the design project (APD) of a new line(s) financed, at least in part, by external funding. Ph. Goudeau (LMP-Poitiers) and O. Thomas (TECSEN-Marseille) accepted be in charge of the general coordination of the groups and synthesis of the collected information.

Contact: Jean Doucet
jean.doucet@synchrotron-soleil.fr

Workshop coordinators: Michèle Sauvage (SOLEIL), Philippe Goudeau (CNRS/AFC), Jean-Lou Lebrun (ENSAM/GFAC-SF2M), Jean Doucet (CNRS)

Control and Data Acquisition technologies

*The mission of the groups in charge of Software & Hardware of Control and Data Acquisition (*Informatique Contrôle Acquisition *and* Électronique Contrôle Acquisition *(ICA and ECA) is to specify, design and develop the data acquisition and control systems of the synchrotron and beamlines. Through the use of generic tools, software and materials, they ensure the coherence of the system as a whole. Thus, they have defined and put in place the standards which constitute today the backbone of the control system.*

What are the control and data acquisition systems at SOLEIL?

The control and data acquisition system of the SOLEIL machine controls the synchrotron equipment and its various components (LINAC, LT1, BOOSTER, LT2, storage ring, front end of the lines, insertions) and data acquisition (experimental results, information on the equipment, etc.)

In the same way, each beamline has its own Control and Data Acquisition system which manages the line equipment and data acquisition.

These systems include all the equipment and software, from electronic signal interfaces to graphic displays.

Systems relying on a distributed architecture with an Ethernet network at its backbone

Control and acquisition devices are numerous, diverse and distributed throughout SOLEIL's infrastructure. In order to simplify their deployment it is preferable that the hardware part and the first software level of the system be located as close as possible to this equipment. As for high level applications and supervision, they involve exchanges of orders and information with, or between, the different elements of the system. That is why a distributed architecture with an Ethernet network at its backbone is being set up.

Systems that require the standardization of software and equipment

Standardizing the products makes the system homogeneous and consistent throughout its architecture, both hardware and software. Therefore, the ICA and ECA groups joined their taskforce to apply this standardization principle to the synchrotron as a whole (machine and beamlines) in order to ensure both homogeneity and consistency.

Standardizing also helps to improve control on the products through the effect of number: for the same requirement, it is always the same solution which is put in place.

Finally, standardization is all the more necessary since the needs are numerous, which is the case with SOLEIL.

Systems that use technologies and products from the industrial field

The construction of SOLEIL is an opportunity to implement recent technologies little used in the field of synchrotron radiation. The initial technical solutions in terms of hardware platforms/systems and operating systems/programming languages will evolve by force of circumstance along the synchrotron lifetime. The solutions retained are therefore innovating, available, non specific and widely adopted by industries to ensure maximum lifetime.

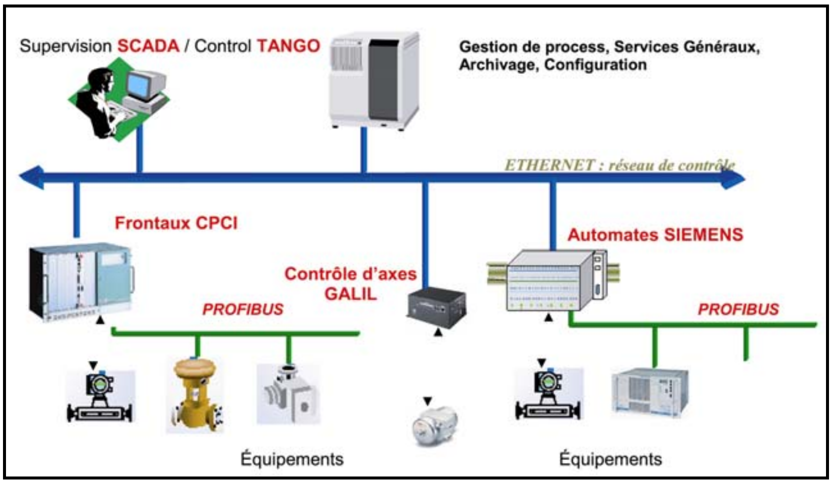


Figure 1 : Architecture du système de contrôle acquisition

Systems born of collaborations with other research centers and SOLEIL

The extent of the work forces to limit the development effort to specificities of the SOLEIL machine. The solution proposed is based, therefore, on solutions already existing on other synchrotrons such as the ESRF.

Collaborations have been set up to take advantage of our strength and experience. SOLEIL's software environment (TANGO) is being developed in partnership with the ESRF and ELETTRA, while part of the equipment used on SOLEIL (CPCI) has been validated by the ESRF in the framework of their machine upgrading.

A control and acquisition system and technological choices validated on the LUCIA * line at the SLS

For the ECA and ICA groups, LUCIA is a real challenge since a life size model is used to validate the technological choices, clear the ground for deploying the system on SOLEIL, and detect upstream the problems and risks that might occur.

LUCIA showed that the control and acquisition system chosen for SOLEIL can adapt to- and fulfil specific needs within reasonable time, i.e., it can be interfaced with the EPICS environment of the SLS undulator.

Finally, the lesson learned from LUCIA highlighted the need for standardizing the equipment and software and developing a partnership spirit between users and the ICA and ECA groups.

By Katy Ho, of the ICA group

In the next Rayon de SOLEIL issue, you will read the follow-up of this article, entitled "A brief outlook on the main software and equipment technologies of the control and acquisition system".

Contact ECA: Pascale Betinelli
pascale.betinelli@synchrotron-soleil.fr

Contact ICA: Alain Buteau
alain.buteau@synchrotron-soleil.fr

Opening of LUCIA, first SOLEIL beamline set up on the Swiss Light Source*

LUCIA was inaugurated on 22 June 2004 on the Swiss synchrotron (Swiss Light Source or SLS) of the Paul Scherrer Institute (PSI). Ralph Eichler and Denis Raoux, directors of the PSI and SOLEIL, respectively, both underlined the great success of this collaboration which started in 2001 between the PSI, LURE, and SOLEIL teams.

Designed to deliver a beam with a diameter in the order of a micrometer in the X-ray intermediary energy field, LUCIA is used in X-ray absorption covering the science of materials (i.e.: studies on the corrosion of concrete), environment (i.e.: studies on contaminated or irradiated soils), geosciences (i.e.: studies on the phenomena of volcanic irruptions), archaeology (i.e.: preservation of antic objects), the analysis of art work and diverse industrial applications, with a stress on spatial resolution.

The Swiss partner (SLS) has installed the undulator, the front end of the line and the infrastructure. The beamline components and the experimental station are supplied by the French partners (LURE and SOLEIL). LUCIA is scheduled to be transferred to SOLEIL by the end of year 2007.
The first proposals should be submitted to the SLS Users Bureau (*Bureau des Utilisateurs*) by September 16. The next program committee will be held in November with allocation of beam time to begin in December 2004. More than 50 French and Swiss research groups are already standing in line for beam time.

*LUCIA : Line for Ultimate Characterizations by Imaging and Absorption

8th Festival of the "Petits Débrouillards" (The resourceful)

The association "Les Petits Débrouillards" (www. lespetitsdebrouillards.org), originally from Quebec, has been active in France since 1984. Its aim is to arouse interest for experimental approach among children and teenagers aged 4 to 16 by giving them the opportunity to discover science and technique through simple and amusing experiments requiring only commonly used or cheap devices.
Among the numerous events organized by this, the most important is its yearly "Festival". The 8th edition of this Festival for the Ile-de-France region was held on the weekend of 19-20 June at the *Cité des Sciences et de l'Industrie*. For the first time an outside guest was invited to participate in the Festival: the Synchrotron SOLEIL.

Models, display panels, educational workshops on the theme of "light" and "light-matter interaction", were held to the sound of music accompanying acrobatics and dance performances that enlivened the Festival. This was an excellent opportunity to address a numerous and motivated audience - 2500 visitors over 2 days - and maybe to create vocations among future researchers (operational for the 4th synchrotron generation!).

Both old and young enjoyed the experiments proposed by the SOLEIL team during the " Festival des Petits Débrouillards" at the *Cité des Sciences et de l'Industrie*.

Ninth edition of the EPAC 5-9 July 2004

Organized this year in Lucerne (Switzerland), the 9th edition of the European conference on particle accelerators gathered 800 participants and 1100 contributions

All types of accelerators were represented: particle physics, nuclear physics and synchrotron radiation. The biggest projects are carried out in the framework of international collaborations. This is the case of the LHC (CERN) for particle physics, currently under construction scheduled to start operation in 2007. The TESLA X-FEL (DESY) project will use linear accelerator technology to produce an X-ray source with a brilliance that could never be achieved with storage rings. These projects include major challenges in the fields of physics and technology. The proton Linacs, which produce high energies (several MW), are very popular since they can be used in various important applications such as the transmutation of nuclear wastes. In the field of radiation, the femto-second impulsion generation is coming of age and the first ideas for generating atto-second impulsions have been reported on.

Moreover, the sub-micronic stability of the position of the beam, the quality of the beam required at the exit of the photo-injectors the difficulties encountered with measurement methods using the beam and the adaptation of theoretical models to physical experience remain problems shared by the physicists which gave rise to very animated discussions.

Contact: Amor Nadji
amor.nadji@synchrotron-soleil.fr

Articles published during the conference are available on line:
http://oraweb03.cern.ch:9000/pls/epac04/toc.htm